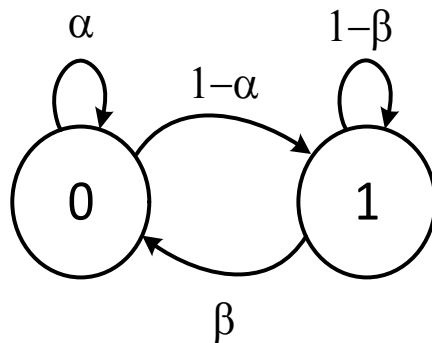


Discrete-time Markov chains

Discrete-time Markov chains

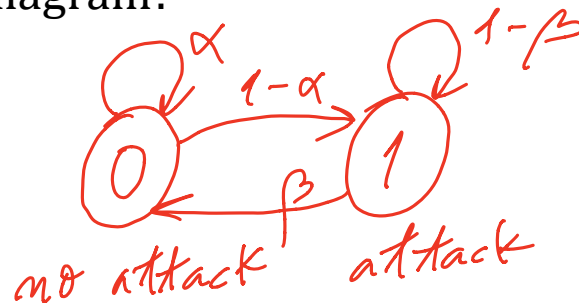
- **Markov chain** - Stochastic process that evolves according to well defined states, where the state “I am going to”, depends only on the state “I am now”, and not on the succession of past states where “I have been”.
- **Discrete-time Markov chain (DTMC)** - Markov chain where the transition between states occurs in discrete steps (e.g. time steps).



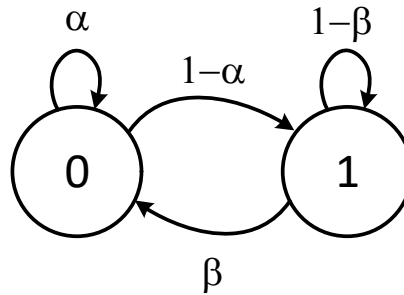
DTMC with 2 states (2-DTMC)

Examples

- Consider a model of daily attacks to a server, where the probability of being attacked tomorrow depends only on whether the server was attacked today. Specifically,
 - if there was no attack today, there will be no attack tomorrow with probability α
 - if there was an attack today, there will be no attack tomorrow with probability β
- Is this a discrete-time Markov chain? If yes, what is the state transition diagram?



Examples



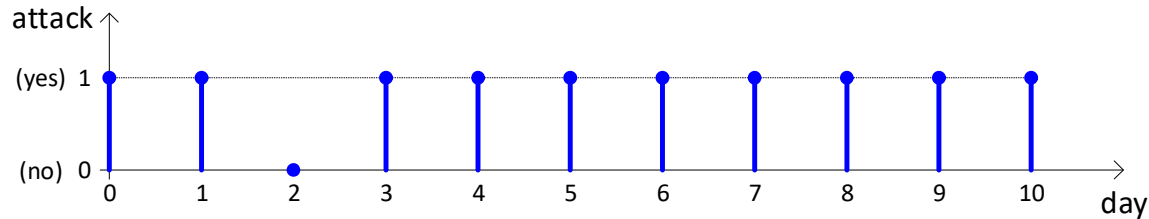
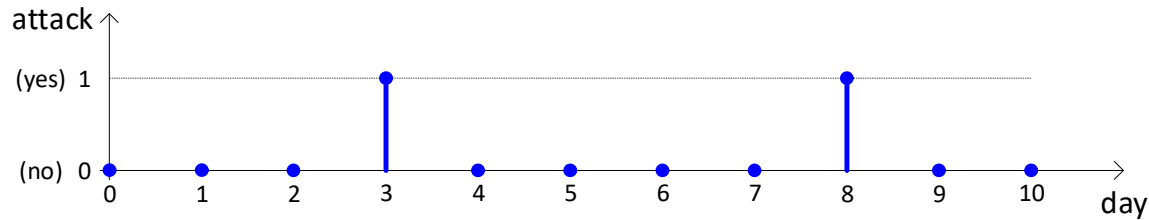
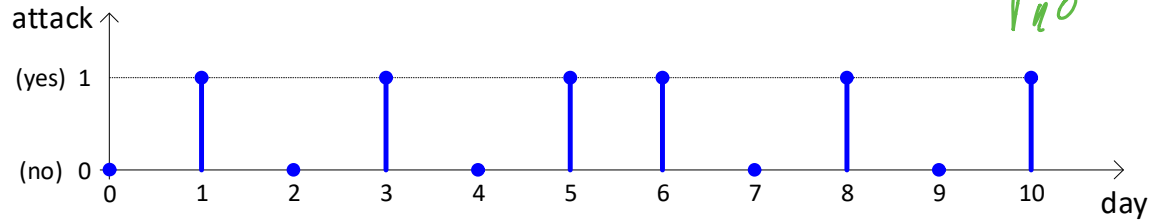
$$P = \begin{bmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{bmatrix}$$

Handwritten green notes: p_{00} , p_{01} , p_{10} , p_{11} with arrows pointing to the corresponding elements in the transition matrix P .

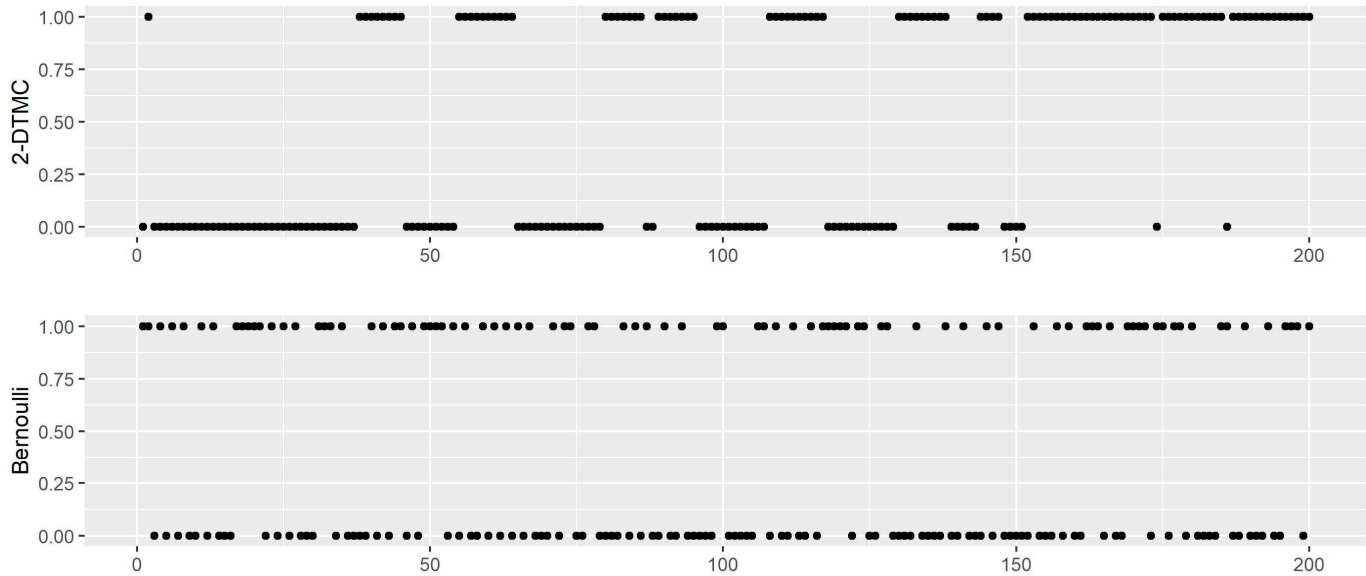
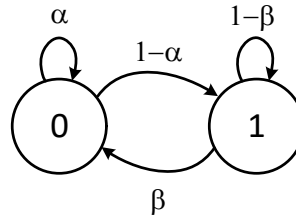
Handwritten red notes: $\alpha \approx 0$, $\beta \approx 1$

Handwritten red notes: $\alpha \approx 1$, $\beta \approx 1$

Handwritten red notes: $\beta \approx 0$, $\alpha \approx 0$



Examples



2-DTMC versus Bernoulli processes
 $\alpha = 0.9, \beta = 0.1, p = (1 - \alpha)/(1 - \alpha + \beta) = 0.5$

Discrete-time Markov chains

- Consider a stochastic process $\{X_n, n = 0, 1, 2, \dots\}$ that can take values from the (countable) set of non-negative integers $\{0, 1, 2, \dots\}$. If $X_n = i$, we say that the process is in state i at time n .
- This stochastic process is a **discrete-time Markov chain (DTMC)**

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = \\ P(X_{n+1} = j | X_n = i) = P_{ij}$$

for all states $i_0, i_1, \dots, i_{n-1}, i, j$ and all $n \geq 0$.

- P_{ij} is called **transition probability**.

Discrete-time Markov chains

- In a discrete-time Markov chain the conditioned distribution of a **future** state X_{n+1} given the **past** states $X_0, X_1, \dots, X_{n-1}$ and the **present** state X_n , is independent of the past states and depends only on the present state.
- The Markov chain is said to be **homogeneous** if P_{ij} is independent of n .
- Since the process has to jump to some state, it results:

$$\sum_{j=0}^{\infty} P_{ij} = 1, \quad i = 0, 1, \dots$$

Discrete-time Markov chains

- The **matrix of transition probabilities** is denoted by **P**

$$\mathbf{P} = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \dots \\ P_{10} & P_{11} & P_{12} & \dots \\ \vdots & \vdots & \vdots & \vdots \\ P_{i0} & P_{i1} & P_{i2} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

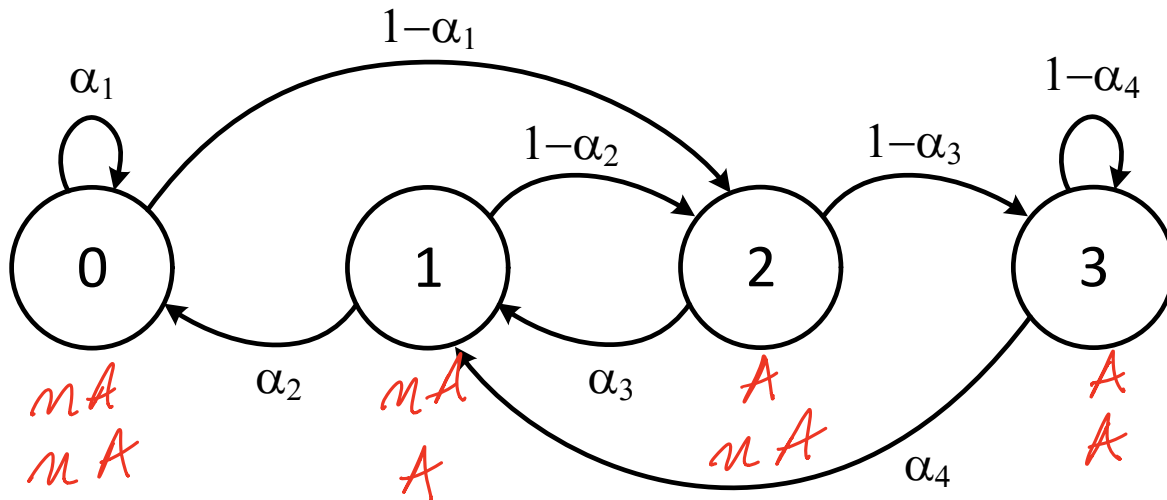
- The sum of row elements is always 1.

Examples

- Suppose now that the probability of being attacked tomorrow depends on whether the server was attacked or not today and yesterday. Specifically,
 - if there was no attack today and yesterday, there will be no attack tomorrow with probability α_1 ;
 - if there was no attack today but an attack yesterday, there will be no attack tomorrow with probability α_2 ;
 - if there was no attack yesterday but an attack today, there will be no attack tomorrow with probability α_3 ;
 - if there was an attack today and yesterday, there will be no attack tomorrow with probability α_4 .
- Is this a discrete-time Markov chain? If yes, what is the state transition diagram?

Examples

*The "today" of the present
will be the "yesterday" of
the future*



Attack example with dependence on two previous days

Chapman-Kolmogorov equations

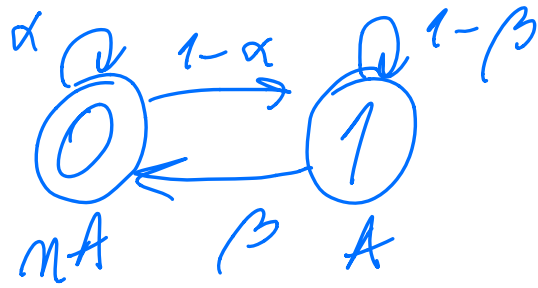
- The n -step transition probability is the probability that a process in state i will be in state j after n transitions ($n \geq 0, i, j \geq 0$):

$$P_{ij}^n = P(X_{n+m} = j | X_m = i)$$

where $P_{ij}^1 = P_{ij}$

- Consider the daily server attack 2-DTMC with $\alpha = 0.1$ and $\beta = 0.6$. Determine the probability that the server is not attacked in three days given that it was not attacked today.

P_{00}^3



$$nA \xrightarrow{\alpha} nA \xrightarrow{\alpha} nA \xrightarrow{\alpha} nA : \alpha^3$$

$$nA \xrightarrow{\alpha} nA \xrightarrow{1-\alpha} A \xrightarrow{\beta} nA : \alpha(1-\alpha)\beta$$

$$nA \xrightarrow{1-\alpha} A \xrightarrow{\beta} nA \xrightarrow{\alpha} nA : \alpha(1-\alpha)\beta$$

$$nA \xrightarrow{1-\alpha} A \xrightarrow{1-\beta} A \xrightarrow{\beta} nA : (1-\alpha)(1-\beta)\beta$$

$$P_{00}^3 = \alpha^3 + 2\alpha(1-\alpha)\beta + (1-\alpha)(1-\beta)\beta$$

$$= 0.325 \quad (\alpha = 0.1, \beta = 0.6)$$

Chapman-Kolmogorov equations

- The **Chapman-Kolmogorov equations** provide a method to calculate the n -step transition probabilities:

$$P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^n P_{kj}^m, \text{ for all } n, m \geq 0, \text{ and all } i, j$$

- The summation has N terms if the Markov chain has N states.
 - $P_{ik}^n P_{kj}^m$ is the probability that, being initially in state i , the process will be in state j after $n + m$ transitions through a path that includes state k in the n -th transition.
- Repeat the previous example using the Chapman-Kolmogorov equations.

$$P_{ij}^{n+m} = \sum_{k=0}^{N-1} P_{ik}^n P_{kj}^m \quad \begin{matrix} N=2 \\ n=1 \\ m=2 \end{matrix}$$

$$P_{00}^3 = P_{00}^{1+2} = \sum_{k=0}^1 P_{0k}^1 P_{k0}^2 =$$

$$= P_{00} P_{00}^2 + P_{01} P_{10}^2$$

$$P_{00}^2 = P_{00}^{1+1} = \sum_{k=0}^1 P_{0k} P_{k0}$$

$$= P_{00} P_{00} + P_{01} P_{10}$$

$$P_{10}^2 = P_{10}^{1+1} = \sum_{k=0}^1 P_{1k} P_{k0}$$

$$= P_{10} P_{00} + P_{11} P_{10}$$

$$\begin{aligned}
P_{00}^3 &= P_{00} P_{00}^2 + P_{01} P_{10}^2 = \\
&= P_{00} (P_{00} P_{00} + P_{01} P_{10}) + \\
&\quad P_{01} (P_{10} P_{00} + P_{11} P_{10}) = \\
&= \frac{\alpha^3}{P_{00} P_{00} P_{00}} + \frac{\alpha(1-\alpha)\beta}{P_{00} P_{01} P_{10}} + \frac{(1-\alpha)\beta\alpha}{P_{01} P_{10} P_{00}} + \\
&\quad + \frac{P_{01} P_{11} P_{10}}{(1-\alpha)(1-\beta)\beta}
\end{aligned}$$

Chapman-Kolmogorov equations

$$N=2: P^{(2)} = \begin{bmatrix} P_{00}^2 & P_{01}^2 \\ P_{10}^2 & P_{11}^2 \end{bmatrix}$$

- The Chapman-Kolmogorov equations can be expressed in matrix form. Denoting the matrix of n -step transition probabilities by $P^{(n)}$ results:

$$P^{(n+m)} = P^{(n)} P^{(m)}$$

and, by induction:

$$P^{(n)} = P^n$$

- Repeat the previous example, now using the matrix form of the Chapman-Kolmogorov equations. In this case:

$$P^{(3)} = P^3 = \begin{bmatrix} \overbrace{0.325}^{P_{00}^3} & \overbrace{0.675}^{P_{01}^3} \\ \underbrace{0.45}_{P_{10}^3} & \underbrace{0.55}_{P_{11}^3} \end{bmatrix}$$

$$p^{(n+m)} = p^{(n)} \cdot p^{(m)}$$

$$\begin{aligned} p^{(2)} &= p^{(1+1)} = p^{(1)} \cdot p^{(1)} = \\ &= p \cdot p = p^2 \end{aligned}$$

$$\begin{aligned} p^{(3)} &= p^{(2+1)} = p^{(2)} \cdot p^{(1)} \\ &= p^2 \cdot p = p^3 \end{aligned}$$

⋮

$$\begin{aligned} p^{(n)} &= p^{(n-1+1)} = p^{(n-1)} \cdot p^{(1)} \\ &= p^{n-1} \cdot p = p^n \end{aligned}$$

$$\rho^{(2)} = \rho^2$$

$$N=2$$

$$= \begin{bmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{bmatrix} \begin{bmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{bmatrix}$$

$$= \begin{bmatrix} \overbrace{\rho_{00}\rho_{00} + \rho_{01}\rho_{10}}^{\rho_{00}^2} & \overbrace{\rho_{00}\rho_{01} + \rho_{01}\rho_{11}}^{\rho_{01}^2} \\ \underbrace{\rho_{10}\rho_{00} + \rho_{11}\rho_{10}}_{\rho_{10}^2} & \underbrace{\rho_{10}\rho_{01} + \rho_{11}\rho_{11}}_{\rho_{11}^2} \end{bmatrix}$$

Transient state probabilities

$$p_{ij}^n \rightarrow \alpha_j^n$$

- The transient state probability is the probability that the process is in state j at time n :

$$\alpha_j^n = P(X_n = j)$$

- Can be obtained by conditioning on the initial state probabilities:

$$\alpha_i = P(X_0 = i) \quad \sum_{i=0}^{N-1} \alpha_i = 1$$

- Specifically

$$\begin{aligned} \alpha_j^n &= \sum_{i=0}^{N-1} P(X_n = j | X_0 = i) P(X_0 = i) \\ &= \sum_{i=0}^{N-1} P_{ij}^n \alpha_i \end{aligned}$$

Transient state probabilities

- In matrix form:

$$\begin{bmatrix} \alpha_0^n & \alpha_1^n & \dots & \alpha_{N-1}^n \end{bmatrix} = \begin{bmatrix} \alpha_0 & \alpha_1 & \dots & \alpha_{N-1} \end{bmatrix} \mathbf{P}^n$$

- In the server attack example, assuming that initially $\alpha_0 = 0.2$ and $\alpha_1 = 0.8$, what is the probability of the server not being attacked after three days? Repeat this exercise for $\alpha_0 = 0.9$ and $\alpha_1 = 0.1$.

- Note that $\mathbf{P}^{(3)} = \mathbf{P}^3 = \begin{bmatrix} 0.325 & 0.675 \\ 0.45 & 0.55 \end{bmatrix}$

$$P^{(3)} = P^3 = \begin{bmatrix} 0.325 & 0.675 \\ 0.45 & 0.55 \end{bmatrix}$$

$$\alpha = 0.1$$

$$\beta = 0.6$$

$$\alpha_0^3 = P_{00}^3 \alpha_0 + P_{10}^3 \alpha_1$$

$$\alpha_0 = 0.2, \alpha_1 = 0.8$$

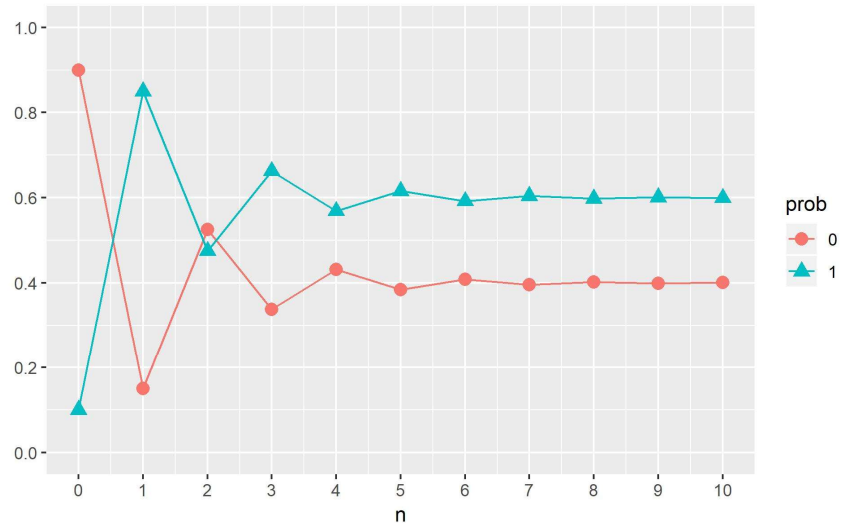
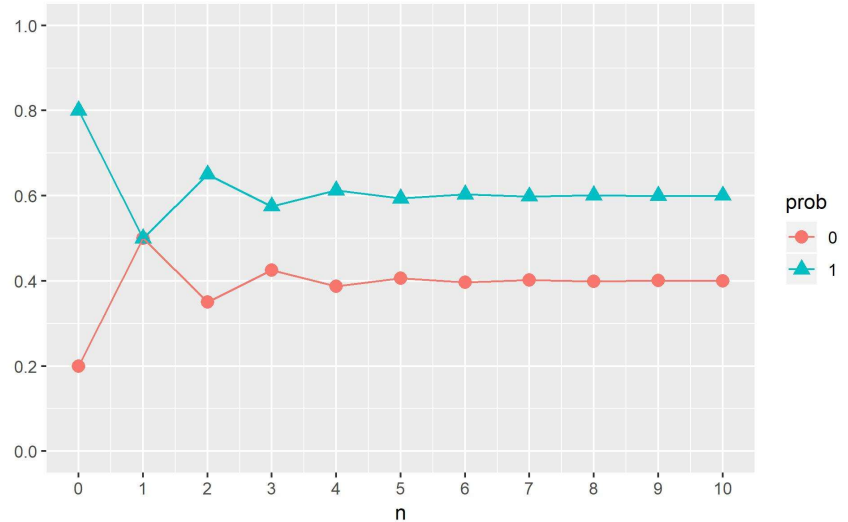
$$\begin{aligned} \alpha_0^3 &= 0.2 \times 0.325 + 0.8 \times 0.45 \\ &= 0.425 \end{aligned}$$

$$\alpha_0 = 0.9, \alpha_1 = 0.1$$

$$\begin{aligned} \alpha_0^3 &= 0.9 \times 0.325 + 0.1 \times 0.45 \\ &= 0.3375 \end{aligned}$$

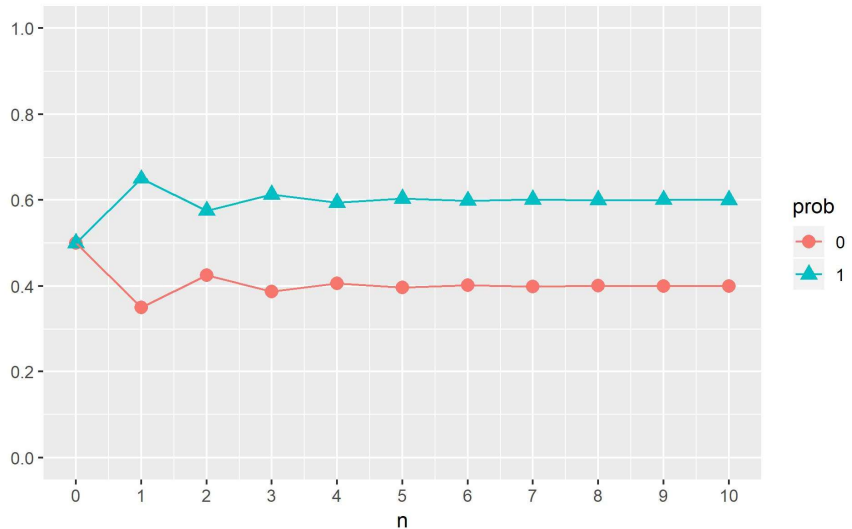
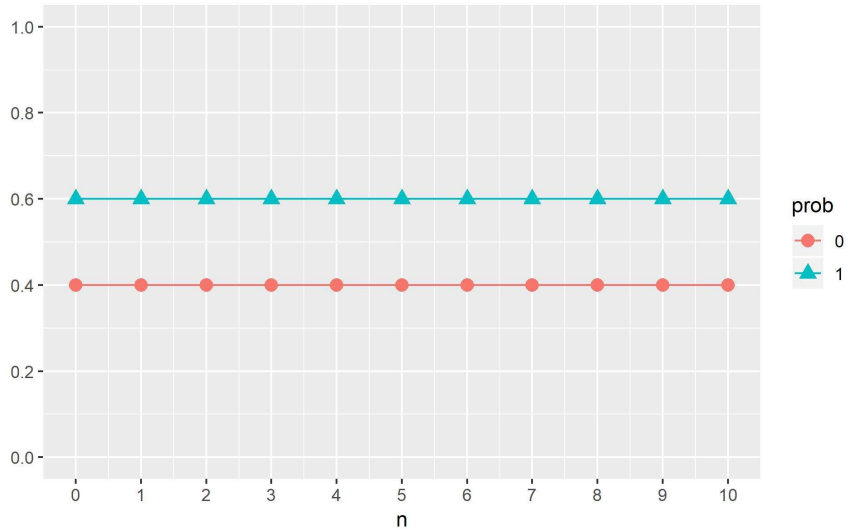
Transient state probabilities

- Values of α_0^n and α_1^n for $n = 0, 1, \dots, 10$, server attack example with $\alpha = 0.1$ and $\beta = 0.6$, and two sets of initial conditions:
 - $(\alpha_0, \alpha_1) = (0.2, 0.8)$
 - $(\alpha_0, \alpha_1) = (0.9, 0.1)$



Transient state probabilities

- Values of α_0^n and α_1^n for $n = 0, 1, \dots, 10$, server attack example with $\alpha = 0.1$ and $\beta = 0.6$, and two sets of initial conditions:
 - $(\alpha_0, \alpha_1) = (0.4, 0.6)$
 - $(\alpha_0, \alpha_1) = (0.5, 0.5)$



Limiting state probabilities

- Limiting state probabilities independent of initial conditions exist if, for each j , P_{ij}^n converges to the same value independently of i , when $n \rightarrow \infty$:

$$\pi_j^L = \lim_{n \rightarrow \infty} P_{ij}^n$$

- Limiting state distribution:

$$\pi^L = [\pi_0^L, \pi_1^L, \dots, \pi_{N-1}^L], \quad \sum_{j=0}^{N-1} \pi_j^L = 1$$

Limiting state probabilities

$$P^{(n)} = P^n$$

$$\pi_j^L = \lim_{n \rightarrow \infty} P_{ij}^n$$

- Can be obtained by multiplying the one-step transition probability matrix by itself a number of times, until the row values are approximately equal $\rightarrow P$

$$P^2 = \begin{bmatrix} 0.55 & 0.45 \\ 0.3 & 0.7 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 0.3831 & 0.6188 \\ 0.4125 & 0.5875 \end{bmatrix}$$

$$P^{13} = \begin{bmatrix} \overbrace{0.3999}^{P_{00}^{13}} & \overbrace{0.6001}^{P_{01}^{13}} \\ \underbrace{0.4000}_{P_{10}^{13}} & \underbrace{0.6000}_{P_{11}^{13}} \end{bmatrix}$$

Stationary state probabilities

- In certain conditions, the limiting state probabilities can be obtained through the **stationary state probabilities**:

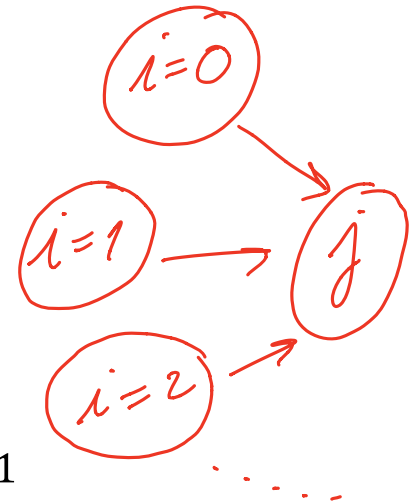
$$\sum_{i=0}^{N-1} \pi_i^S P_{ij} = \pi_j^S, \quad j = 0, 1, \dots, N-1$$

$$\sum_{i=0}^{N-1} \pi_i^S = 1$$

Stationary state probabilities

- The left-hand side of

$$\sum_{i=0}^{N-1} \pi_i^S P_{ij} = \pi_j^S, \quad j = 0, 1, \dots, N-1$$



represents the probability of being in state j one transition from now, given that the current distribution of states is $[\pi_0^S, \pi_1^S, \dots, \pi_{N-1}^S]$. This probability equals precisely the probability of state j (the right-hand side).

- If we start with some probability distribution $[\pi_0^S, \pi_1^S, \dots, \pi_{N-1}^S]$, one step later we will have the same distribution.

Stationary state probabilities

- In matrix form

$$\pi^S = \pi^S \mathbf{P} \quad \mathbf{1} = \pi^S \mathbf{E}$$

where $\mathbf{1}$ is a row vector with N elements equal to one and $\mathbf{E}$ is an $N \times N$ matrix with all elements equal to one.

- Using this form:

$$\pi^S = \mathbf{1}[\mathbf{P} + \mathbf{E} - \mathbf{I}]^{-1}$$

- In the server attack example with $\alpha = 0.1$ and $\beta = 0.6$, determine the stationary state probabilities of being attacked and not being attacked.

$$\pi = \pi P$$

$$I = \pi E$$

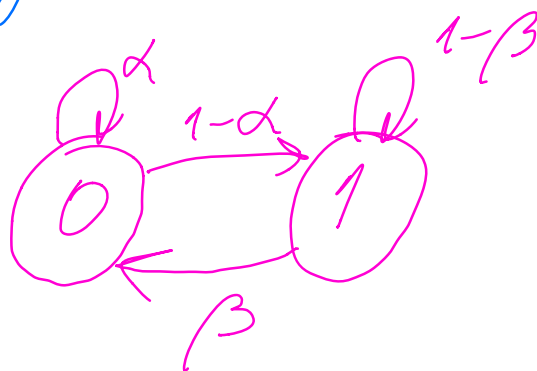
$$\pi + I = \pi P + \pi E$$

$$I = \pi (P + E - I)$$

$$\pi = I (P + E - I)^{-1}$$

$$\pi_j = \sum_{i=0}^{N-1} \pi_i p_{ij}, \quad j = 0, 1, \dots, N-1$$

$$\sum_{i=0}^{N-1} \pi_i = 1$$



$$\pi_0 + \pi_1 = 1$$

$$j=0: \pi_0 = \pi_0 \overbrace{p_{00}}^{\alpha} + \pi_1 \overbrace{p_{10}}^{\beta}$$

$$j=1: \pi_1 = \pi_0 \overbrace{p_{01}}^{1-\alpha} + \pi_1 \overbrace{p_{11}}^{1-\beta}$$

$$\begin{cases} \pi_0 = \pi_0 \alpha + \pi_1 \beta \\ \pi_0 + \pi_1 = 1 \end{cases}$$

$$\begin{cases} \pi_0 = \pi_0 \alpha + \pi_1 \beta \\ \pi_0 + \pi_1 = 1 \end{cases}$$

$$\pi_0 = \pi_0 \alpha + (1 - \pi_0) \beta$$

$$\pi_0 - \pi_0 \alpha + \pi_0 \beta = \beta$$

$$\pi_0 = \frac{\beta}{1 - \alpha + \beta}$$

$$\pi_1 = 1 - \pi_0 = \frac{1 - \alpha + \beta - \beta}{1 - \alpha + \beta}$$

$$= \frac{1 - \alpha}{1 - \alpha + \beta}$$

$$\pi_0 = \frac{\beta}{1 - \alpha + \beta}$$

$$\pi_1 = \frac{1 - \alpha}{1 - \alpha + \beta}$$

$$\begin{array}{ll} \alpha = 0.4 & \pi_0 = 0.4 \\ \beta = 0.6 & \pi_1 = 0.6 \end{array}$$

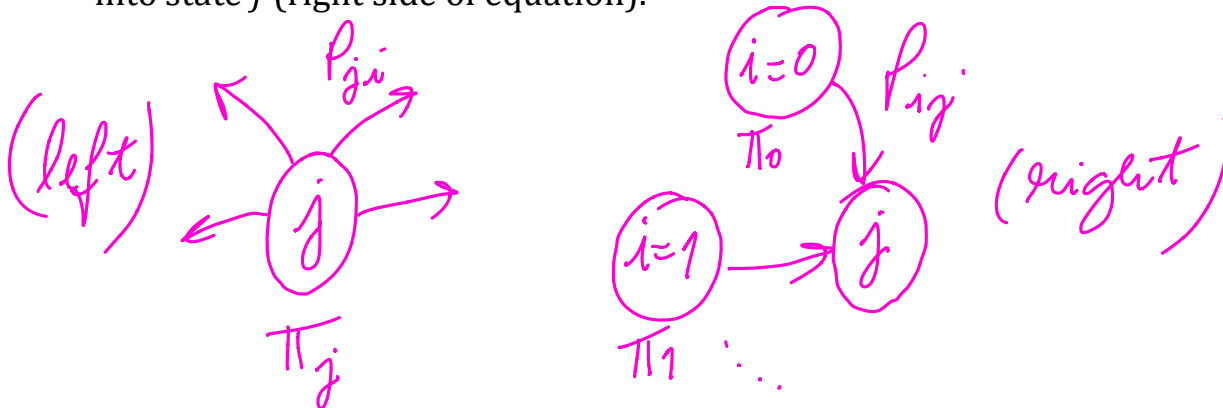
$$\alpha = \beta = 0.5 \quad \pi_0 = \pi_1 = 0.5$$

Global balance equations

- Global balance equations ($j \geq 0$):

$$\pi_j^S \sum_i P_{ji} = \sum_i \pi_i^S P_{ij}$$

- $\pi_i^S P_{ij}$ can be interpreted as the frequency of transitions from i to j .
- Thus, the global balance equations state that the frequency of transitions out of state j (left side of equation) equals the frequency of transitions into state j (right side of equation).



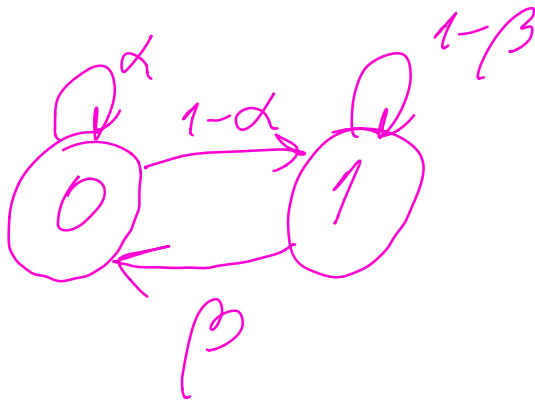
$$\sum_{i=0}^{\infty} p_{ji} = 1$$

$\times \pi_j$

$$\pi_j \sum_{i=0}^{\infty} p_{ji} = \pi_j$$

$$\pi_j = \sum_{i=0}^{\infty} \pi_i p_{ij}$$

$$\pi_j \sum_{i=0}^{\infty} p_{ji} = \sum_{i=0}^{\infty} \pi_i p_{ij}$$



$$\cancel{\pi_0 \alpha} + \pi_0(1-\alpha) = \cancel{\pi_0 \alpha} + \pi_1 \beta$$

$$\pi_0(1-\alpha) = \pi_1 \beta$$

$$\begin{cases} \pi_1 \beta = \pi_0(1-\alpha) \\ \pi_0 + \pi_1 = 1 \end{cases}$$

$$(1 - \pi_0) / \beta = \pi_0(1 - \alpha)$$

$$\pi_0(1 - \alpha + \beta) = \beta \quad \pi_0 = \frac{\beta}{1 - \alpha + \beta}$$

$$\pi_1 = 1 - \pi_0 = \frac{1 - \alpha}{1 - \alpha + \beta}$$

Existence of limiting state probabilities

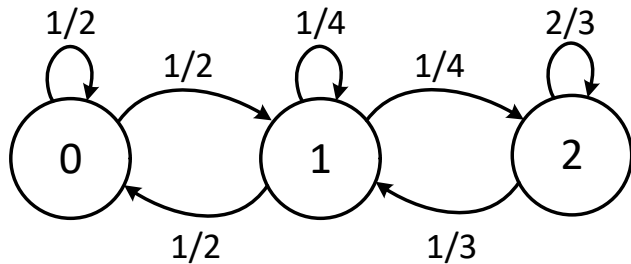
- If the limiting distribution exists, then the limiting distribution is a stationary distribution and no other stationary distribution exists
- **Finite-state DTMCs**: limiting distribution exists (with strictly positive probabilities) if the chain is **irreducible** and **aperiodic**.
- Infinite-state DTMCs further require that the chain be **positive recurrent**. Irreducible, aperiodic, and positive recurrent chains are also called **ergodic**.

Limiting state probabilities interpreted as rates

- The limiting state probability π_j represents also the proportion of time that the process spends in state j .

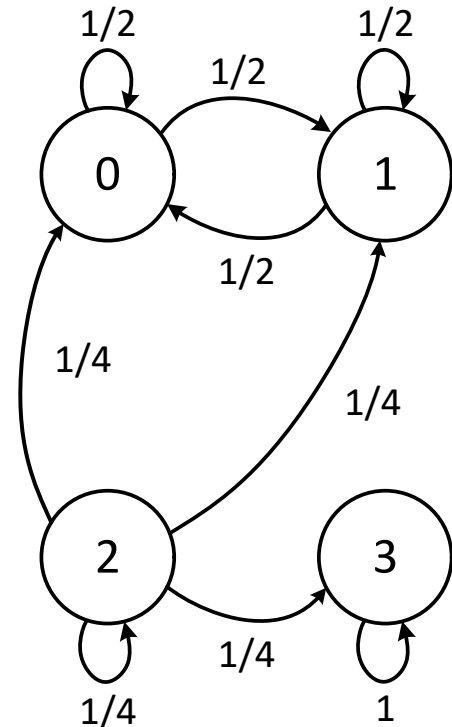
State classification

- A Markov chain is **irreducible** if all states communicate with each other.



irreducible

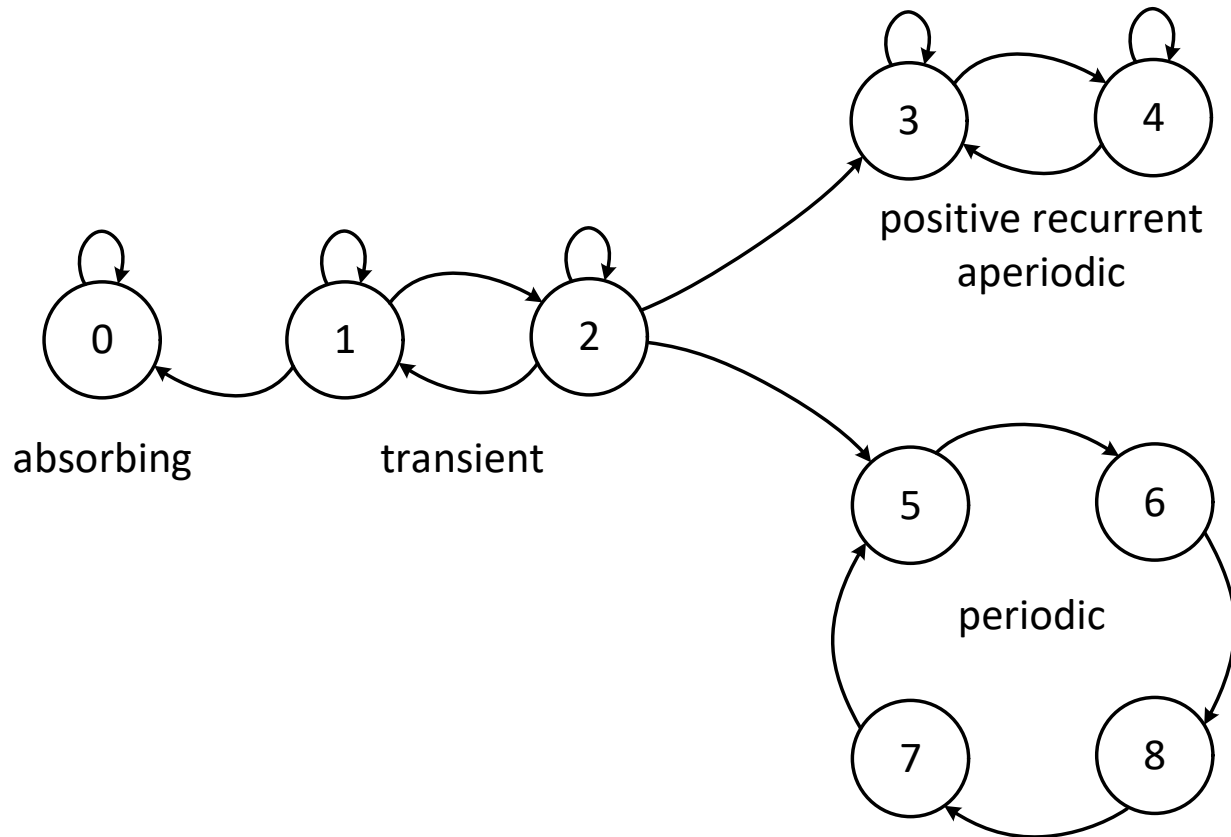
not irreducible



State classification

- A state is **recurrent** if the probability of ever returning to the state is 1; otherwise, the state is **transient**.
- A recurrent state is visited infinitely often. Given a recurrent state, if the mean recurrence time is finite, the state is called **positive recurrent**; otherwise it is called **null recurrent**.
- A state is **periodic** if the chain can only return to the state after a multiple of T steps, where $T > 1$.
- Positive recurrent and aperiodic states are called **ergodic**. A Markov chain is said ergodic if all its states are ergodic.

State classification

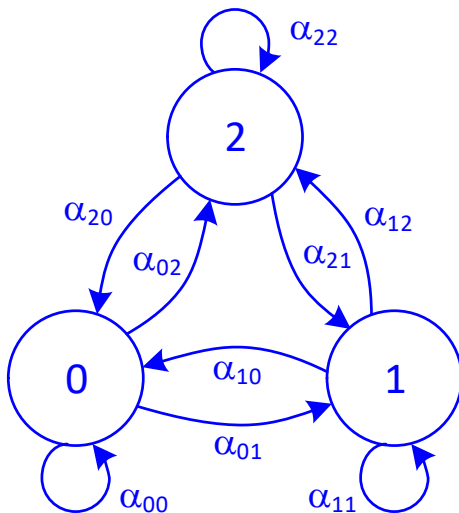


Estimation of transition probabilities

- Can be estimated by the **relative frequency of transitions counts**

- 3-DTMC example:

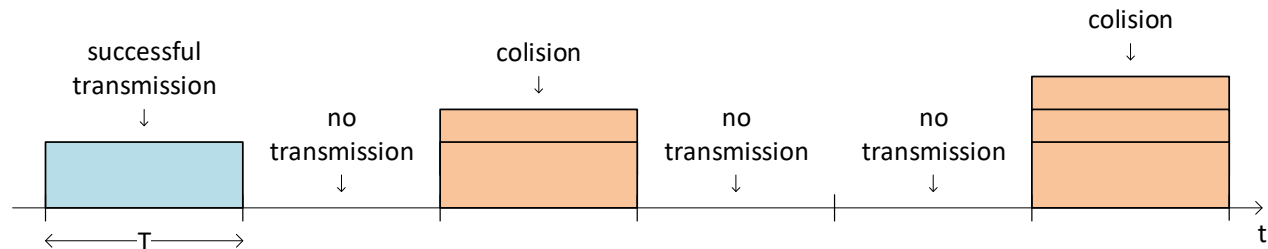
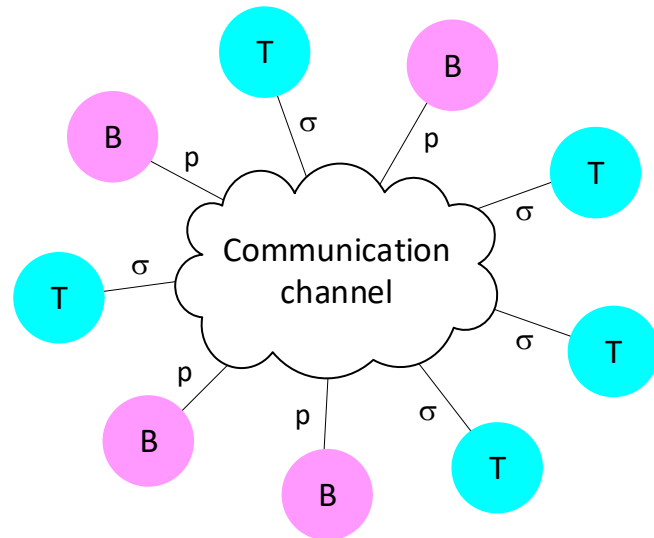
data: 0 1 0 2 2 0 1 1 2 0 2 0 0 0 2 0 1 1 2 1 2 1 0 2



$$\text{transition counts} = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 2 & 3 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\text{relative frequencies} = \begin{bmatrix} \overbrace{2/9}^{\alpha_{00}} & \overbrace{3/9}^{\alpha_{01}} & \overbrace{4/9}^{\alpha_{02}} \\ \underbrace{1/7}_{\alpha_{20}} & \underbrace{2/7}_{\alpha_{21}} & \underbrace{3/7}_{\alpha_{22}} \\ \underbrace{4/7}_{\alpha_{20}} & \underbrace{2/7}_{\alpha_{21}} & \underbrace{1/7}_{\alpha_{22}} \end{bmatrix}$$

Performance of slotted ALOHA



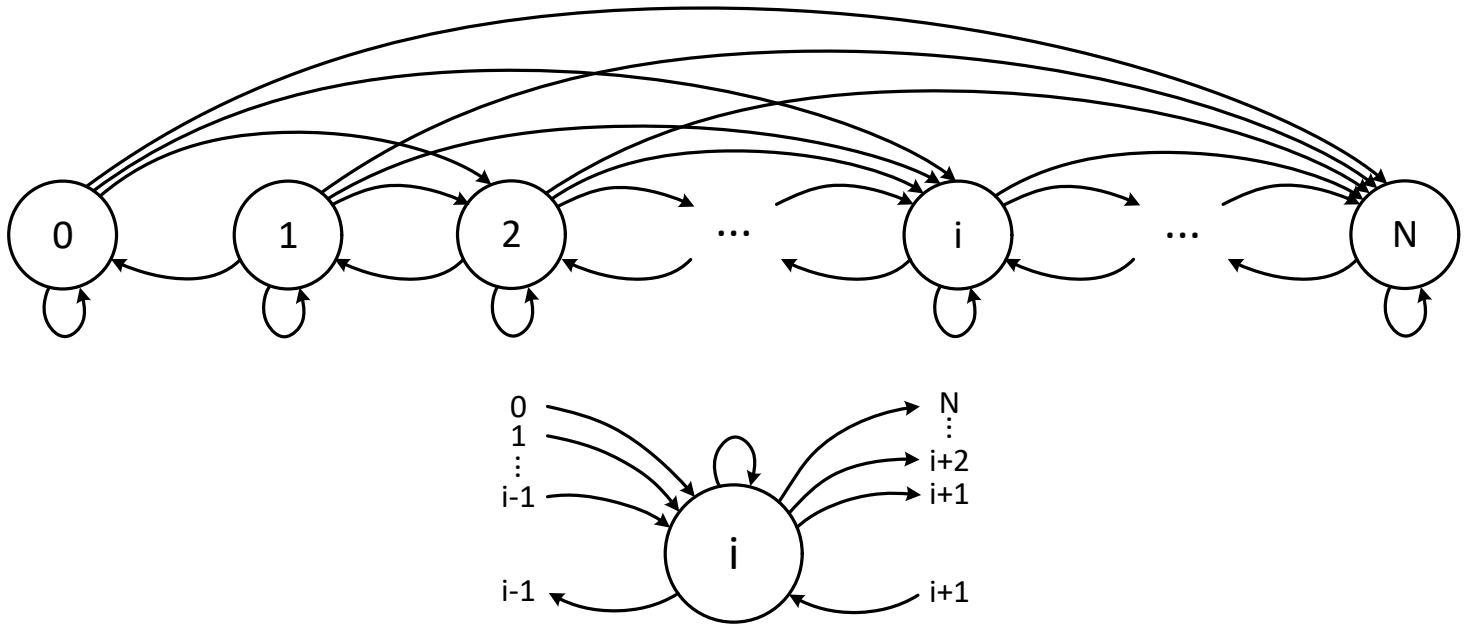
Performance of slotted ALOHA

* throughput $\equiv$ proportion of time that the channel is used with successful transmissions

- A simple model:
 - There are N users (a finite number)
 - Time is slotted and the duration of one slot is the time to transmit a packet.
 - Each user can be in one of two states: THINKING or BACKLOGGED.
 - THINKING users transmit one packet in a slot with probability σ and change to the BACKLOGGED state if they collide.
 - BACKLOGGED users have exactly one packet waiting and transmit in a slot with probability p .
 - If two or more users transmit in the same slot there will be a collision; collided users are placed in the BACKLOGGED state (if not already there).
 - If only one user transmits in a slot, the transmission will have success and the user returns to the THINKING state (if not already there).
- The goal is to calculate the mean throughput of the ALOHA channel

Performance of slotted ALOHA

*States $\equiv$ number
of backlogged
users*



State transition diagram

* i backlogged users
 * $N-i$ thinking users

Performance of slotted ALOHA

- States of Markov Chain = Number of BACKLOGGED users
- $j < i - 1$, only one successful transmission in a time slot is possible:

backlog cannot decrease by more than one $\rightarrow p_{ij} = 0 \rightarrow \text{e.g. } p_{31} = 0$

- $j = i - 1$ (backlog decreases by one), exactly one transmission from BACKLOGGED users AND no transmissions from THINKING users:

$$p_{ij} = \underbrace{ip(1-p)^{i-1}}_{1 \text{ tx of } B} \underbrace{(1-\sigma)^{N-i}}_{0 \text{ tx of } T}$$

- $j = i$, no transmissions from THINKING users AND zero or more than one transmissions from BACKLOGGED users OR exactly one transmission from THINKING users AND no transmissions from BACKLOGGED users:

$$p_{ij} = \underbrace{(1-\sigma)^{N-i}}_{0 \text{ tx of } T} \underbrace{[1 - ip(1-p)^{i-1}]}_{1 \text{ tx of } T} \rightarrow 0 \text{ or } > 1 \text{ tx of } B$$

$$+ \underbrace{(N-i)\sigma(1-\sigma)^{N-i-1}}_{1 \text{ tx of } T} \underbrace{(1-p)^i}_{0 \text{ tx of } B}$$

Performance of slotted ALOHA

- $j = i + 1$ (backlog increase by one), exactly one transmission from THINKING users AND one or more transmissions from BACKLOGGED users:

$$p_{ij} = \underbrace{(N - i)\sigma(1 - \sigma)^{N-i-1}}_{1 \text{ tx of } T} \underbrace{(1 - (1 - p)^i)}_{> 0 \text{ tx of } B}$$

- $j > i + 1$ (backlog increases by $j - i$), $j - i$ transmissions from THINKING users :

$$p_{ij} = \underbrace{\binom{N-i}{j-i} \sigma^{j-i} (1 - \sigma)^{N-j}}_{(j-i) \text{ tx of } T}$$

→ The transmissions by each type of user (T or B) can be described by a

Binomial distribution.

We want to determine the probability of a number of transmissions from a number of T or B users.

$$\text{Recall: } P\left(\begin{matrix} i \text{ successes} \\ n \text{ trials} \end{matrix}\right) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$\binom{n}{i} = \frac{n!}{i!(n-i)!}$$

$$P\left(\begin{matrix} 1 \text{ success} \\ n \text{ trials} \end{matrix}\right) = \binom{n}{1} p (1-p)^{n-1} = np(1-p)^{n-1}$$

$$P\left(\begin{matrix} 0 \text{ successes} \\ n \text{ trials} \end{matrix}\right) = \binom{n}{0} p^0 (1-p)^n = (1-p)^n$$

Performance of slotted ALOHA

- Conditional throughput:

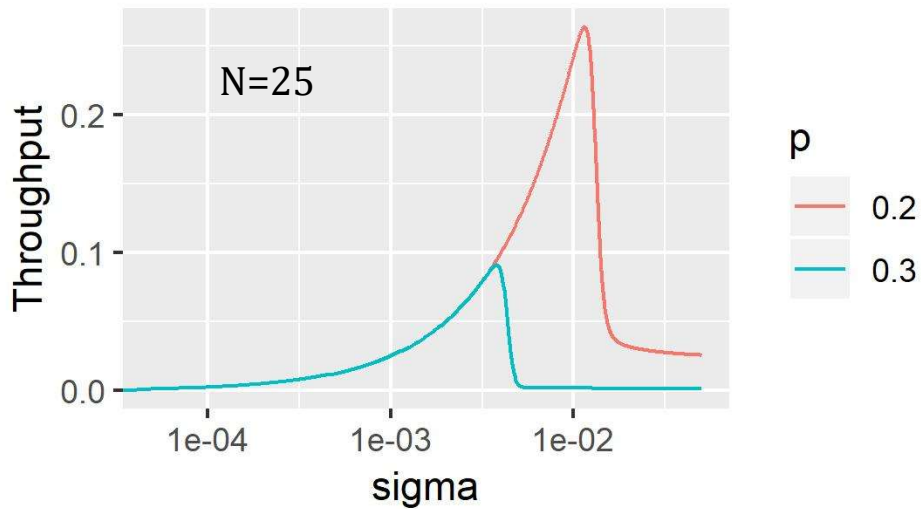
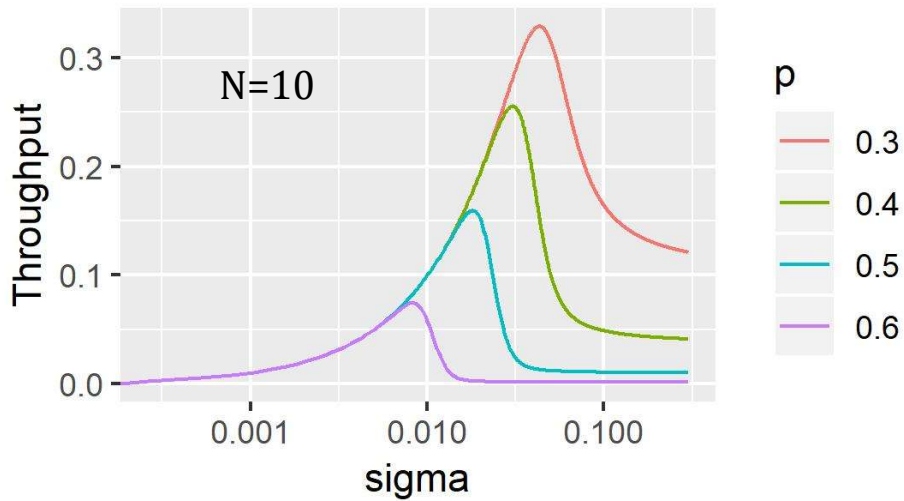
$$P_{suc}(i) = Prob(\text{Successful slot} | i \text{ users backlogged})$$

$$= \underbrace{(N-i)\sigma(1-\sigma)^{N-i-1}}_{1 \text{ tx of } T} \underbrace{(1-p)^i}_{0 \text{ tx of } B} + \underbrace{ip(1-p)^{i-1}}_{1 \text{ tx of } B} \underbrace{(1-\sigma)^{N-i}}_{0 \text{ tx of } T}$$

- Throughput:

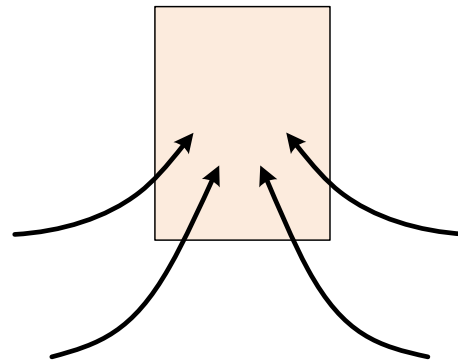
$$S = \sum_{i=0}^N P_{suc}(i) \pi_i$$

Performance of slotted ALOHA



Google's PageRank algorithm

- Problem: **How to rank Web pages?**
- **Citation counting:** count the number of links to a page (often called backlinks); page with the highest count ranked first
 - Problem 1: Not all links are equal; a link to my page from the Yahoo Web page is more important than a link from Rui Valadas' Web page
 - Problem 2: Easily tricked; to become important, I just create thousands of Web pages pointing to my Web page



Google's PageRank algorithm

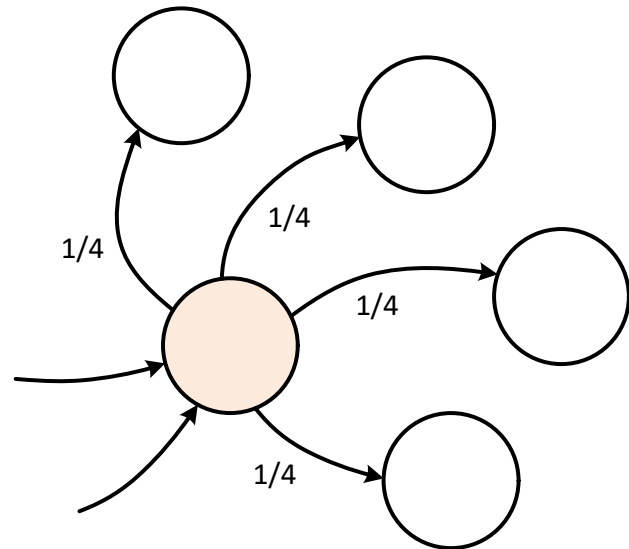
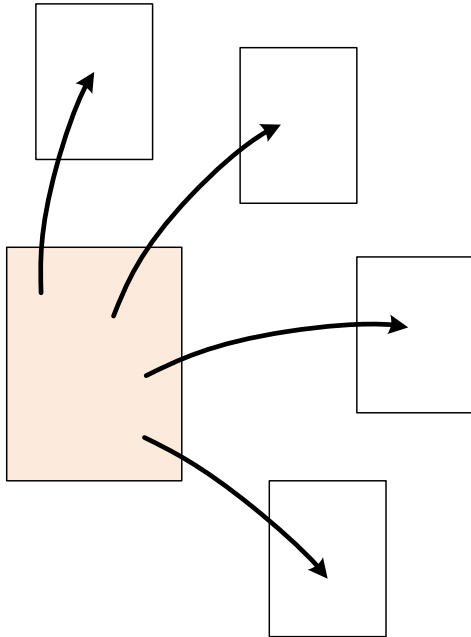
- Possible alternative: weight each ~~pointer~~^{link} by the number of pages pointing to it^{is originating page}
 - Problem: Easily tricked; to become important, I just create thousands of Web pages pointing to each other as well as to my Web page
- Google's solution: define page rank recursively; page has high rank if the sum of the ranks of its backlinks is high. Use:

$$\pi_j = \sum_{i=1}^n \pi_i P_{ij}$$

Google's PageRank algorithm

- Google's PageRank algorithm
 - Create a DTMC where there is one state for each Web page and there is a transition from state i to state j if page i has a link to page j
 - If page i has $k > 0$ outgoing links, then set the probability of each outgoing arrow from state i to be $1/k$.
 - Solve the DTMC to determine the ~~limiting~~ probabilities. Pages are then ranked based on the ~~limiting~~ probabilities (higher probability first).
stationary state
- This simple algorithm was the original basis behind the entire Google company!
stationary state
- Practical considerations
 - Number of states is huge
 - Google solution: limiting probabilities are obtained taking powers of P , and many other tricks...

Google's PageRank algorithm



The Web is a large DTMC